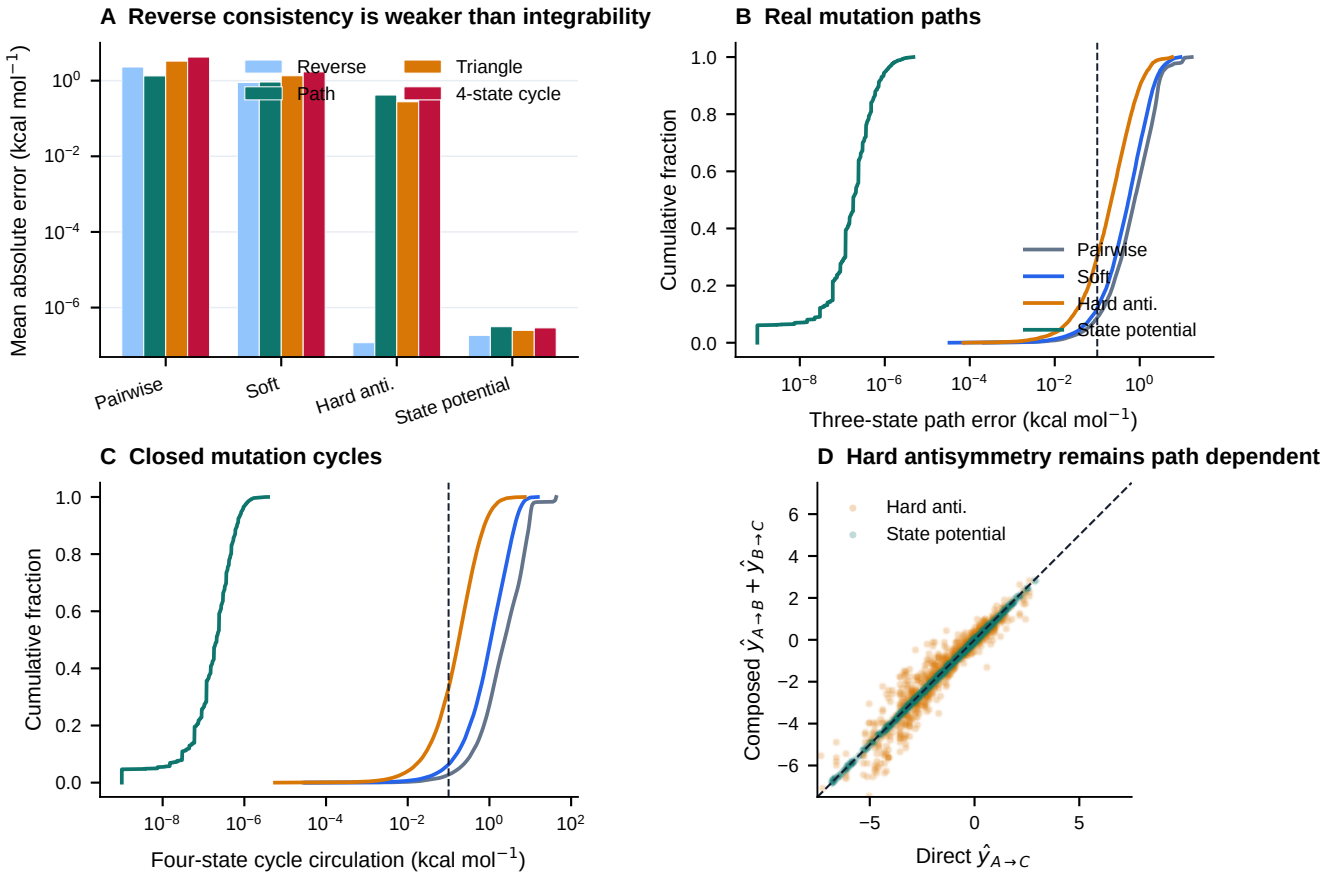
A Reverse consistency is weaker than integrability
Mean absolute error (kcal mol^{-1})
10^{0}
10^{-2}
10^{-4}
10^{-6}
Reverse
Path
Triangle
4-state cycle
Pairwise
Soft
Hard anti.
State potential
B Real mutation paths
Cumulative fraction
1.0
0.8
0.6
0.4
0.2
0.0
Pairwise
Soft
Hard anti.
State potential
10^{-8}
10^{-6}
10^{-4}
10^{-2}
10^{0}
Three-state path error (kcal mol^{-1})
C Closed mutation cycles
Cumulative fraction
1.0
0.8
0.6
0.4
0.2
0.0
10^{-8}
10^{-6}
10^{-4}
10^{-2}
10^{0}
10^{2}
Four-state cycle circulation (kcal mol^{-1})
D Hard antisymmetry remains path dependent
Hard anti.
State potential
Composed $\hat{y}_{A\to B} + \hat{y}_{B\to C}$
6
4
2
0
−2
−4
−6
−5
0
5
Direct $\hat{y}_{A\to C}$